

Cartoon Sliding: An MR System for Experiencing Sliding Down a Cliff

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ABSTRACT

We present an MR system for experiencing the cartoon-like event of sliding down a cliff and stopping by sticking a knife, with enhancement of tactile and muscle stimuli. A player grabs a knife device and presses it against a wall device that rotates upwards in contrast to the player who virtually slides down. We demonstrate an MR content in which a player experiences sliding down the virtual cliff while handling the knife device on the rotating wall device.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Mixed reality, cartoon physics, sliding, cliff, wall, knife, interface

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1 INTRODUCTION

Research on mixed reality (MR) have recently progressed that employs not only audiovisual information but also other sensory and environmental information in the real world. In order to realize such MR experiences, virtual and real geometric/physical conditions on objects have to be synchronized in a consistent way. As for audiovisual information, products such as Microsoft's HoloLens have enabled immersive MR experiences with strong sense of presence. As for tactile information, systems such as Grabity [Choi et al. 2017] have enabled physically consistent feel of touching and manipulating virtual objects. However, more unrealistic or fantastic events that are related to "cartoon physics" have not yet been fully explored in the MR environment with tactile and muscle stimuli.

We develop an MR system to realize the cartoon-like event of sliding down a cliff and stopping by pushing a knife against the wall, with physical engagement of especially the upper body. In our system, a player with a head-mounted display (HMD) grabs a knife device and tries to insert it to the wall device that slides *upwards* as the player falls; thus, relatively experiencing the fall.

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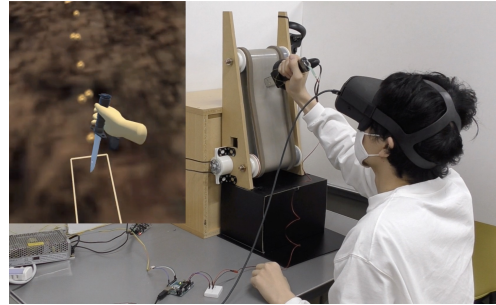


Figure 1: A player experiencing Cartoon Sliding.

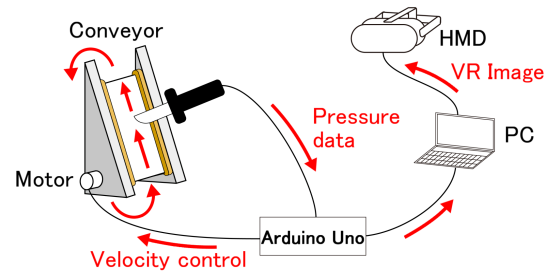


Figure 2: System overview.

2 SYSTEM

In order to realize the tactile and muscle stimuli of sliding down a cliff, especially at our arms without physically falling, we developed a rotatable wall device and a knife device. The system mainly consists of these two devices, HMD (Oculus Quest), Arduino Uno, and a computer. The overview of the system is shown in Figure 2 and some details of the system are described below.

2.1 Wall Device

The wall device is shown in Figure 3 on the left. The device consists of a conveyor with a roll of belt of size 80.0×15.0 [cm], two pulleys of diameter 7.0 [cm], a drive roller of diameter $D = 3.0$ [cm], and a DC motor (YM-2774, 6-12 [V], 2.6 [A], max 15,520 [rpm] without load). The slope of the wall is about 80 degrees. The motor speed is up to around $N=1,000$ [rpm] because of the load of the conveyor. The upper velocity of the conveyor is estimated as $(\pi D/100) \cdot (N/60) = 1.57$ [m/s]. We note that this value is far less than the actual speed of the physical falling, but it works for the MR experience. The velocity of the conveyor is controlled by Arduino Uno using PWM in 256 levels.



Figure 3: The wall device and the knife device.

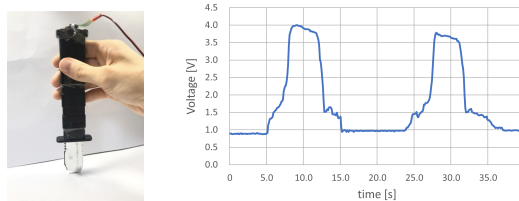


Figure 4: The pressed knife and voltage during the action.

2.2 Knife Device

For the knife device, we use a gimmick knife with a blade that can be retracted into the handle when pressed, as shown in Figure 3 on the right, because this type of knife is pressed against a surface just like the real one, which is desirable in our MR setting. At the bottom of the handle, a pressure sensor is attached to obtain pressure data represented by voltage. The pressed knife and an example of voltage during the retraction are shown in Figure 4. The voltage data are sent to the wall device to control the velocity of the conveyor as well as to the computer to generate VR image.

2.3 Applications

First, Arduino Uno controls the velocity of the conveyor using PWM in 256 levels, where 255 corresponds to the maximum velocity. We have used the values up to 100 to avoid the slippage of the pulley belt. In the beginning, the velocity is increased, for example, by 5 at every 0.3 sec, achieving 100 in 6 sec (we can also realize quadratic speed-up, which corresponds to free fall). After that, we increment by 1 at every time frame (1 frame = 0.1 sec) for expressing the increase of the sliding speed. When Arduino receives positive pressure data, we reduce some value, say, proportional to the pressure data, from the present value of PWM, resulting in slowing down the conveyor.

We also developed VR application in which a player slides down a virtual cliff with velocity determined by the pressure data. The system was developed with 2020.3.10f1. The controllers of Oculus were attached to the wall and knife devices to track their positions.

3 DEMONSTRATION

The MR experience proceeds as follows. A player wearing HMD grabs the knife device and faces the wall device. When the system is started, the conveyor speeds up in 6 sec this time (Figure 5 (a)). When the knife is pressed against the conveyor belt and the blade retracts into the handle, the conveyor slows down based on the procedure outlined in Section 2.3.1. At the same time, it is slowed down in the virtual image displayed by HMD (Figure 5 (b)). When

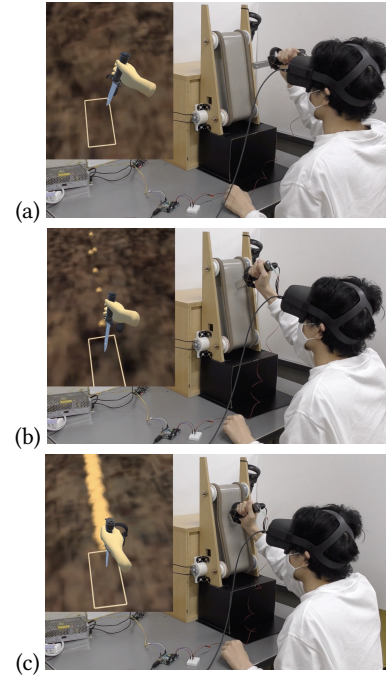


Figure 5: Demonstration of the MR sliding experience.

the knife is pressed harder, the conveyor finally stops and the player also stops sliding down both virtually and physically (Figure 5 (c)).

4 DISCUSSION

For the cartoon-like event of sliding down a cliff and stopping, we confirmed that a part of the whole body experience, especially, the one around the arm is physically realizable. Furthermore, we noticed through the experience of the demonstration that the task of physically pressing the knife against the moving wall has a unique sense of challenge, enjoyment, and accomplishment, which can not be appreciated by just watching a cartoon or animation, which should imply a potential of mixed reality.

On the other hand, since the system mainly focuses on the upper body, the lower body tends to be stable, which suppresses the sense of falling. We have tested a cushion around the knees to help the body to be slightly unstable. We plan to extend the system for further destabilization. Another problem is on the softness of the conveyor belt, which prevents us from fully immersing ourselves in the situation. We should improve the physical properties of the belt, including the texture. Finally, applying the ideas of the system to other cartoon-like events should be also among future work.

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